

JEE MAIN REVISION MASTER: THERMODYNAMICS

2026 3 QUESTIONS 5% WEIGHTAGE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 14 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Kcet Thermodynamics - Core Application	14 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Thermodynamics - Core Application (High Yield — 14 PYQs)

Core Principle: Key problem pattern in Kcet Thermodynamics.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "A gas can be taken from A to B via two different paths ACB and ADB. When path ACB is used, 60 J of heat flows into the system and 30 J o..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "Which of the following is a correct statement for a thermodynamic system?..." [Ans: N/A]
- [2026 Shift] Q#3 (MCQ): "A: Entropy of a perfect crystalline solid at absolute zero approaches zero. B: For spontaneity of a reaction, $T\Delta S > \Delta H$. Among t..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

A gas can be taken from A to B via two different paths ACB and ADB. When path ACB is used, 60 J of heat flows into the system and 30 J of work is done by the system. If path ADB is used, work done by the system is 10 J. The heat flow into the system in path ADB is

- (A) 80 J
- (B) 20 J
- (C) 100 J
- (D) 40 J

Correct Answer: None

Solution Analysis: By the first law of thermodynamics, $\Delta U = Q - W$. Here, Q is the heat supplied to the system, and W is the work done by the system. For path ACB: Heat added, $Q_{(ACB)} = 60$ J Work done, $W_{(ACB)} = 30$ J So, $\Delta U = 60 - 30 = 30$ J The change in internal energy depends only on the initial state A and final state B. It does not depend on the path followed. So, for path ADB also, $\Delta U = 30$ J For path ADB: Work done, $W_{(ADB)} = 10$ J Using the first law again, $\Delta U = Q_{(ADB)} - W_{(ADB)}$
 $30 = Q_{(ADB)} - 10$ $Q_{(ADB)} = 30 + 10 = 40$ J The heat flow into the system in path ADB is 40 J.